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A physics-informed stochastic model for the long-term correlation of earthquakes

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Simone Barani[1], Matteo Taroni[2] ✉, Davide Zaccagnino[3], Pietro Artale Harris[2] & Giuseppe Petrillo[4]

In this work, a simple physics-informed stochastic earthquake catalog simulator is presented for the modeling of long memory on a single fault (or crustal volume). The simulator combines the fundamental statistical laws of seismology (i.e., the Gutenberg-Richter law and the exponential distribution of inter-event times of spontaneous events) with two physical assumptions: 1) earthquake magnitudes are constrained by the amount of potential energy accumulated within the crustal volume at a given time, 2) existence of an upper boundary for stress on the fault. The consequence is a deviation from the conventional exponential inter-event distribution. We analyze a set of simulated catalogs to inspect long memory. Our results show that, depending on the relative proportion of energy accumulation and release, catalogs can exhibit complex patterns and long memory behavior. Despite its simplicity, our simulator can replicate the key features of seismic activity observed in real data, thus enabling a consistent physical interpretation.

Although the concept of long memory, or long-range dependence (also long-term correlation or persistence), in the seismic process dates back to the late 80s[1], awareness of this has grown in recent years thanks to the increasing completeness of earthquake data and the extent and quality of parametric and paleoseismic recordings, which has allowed the application of various statistical techniques aimed at evaluating correlations among observables of earthquake time series[2–9]. Research on this topic was originally fueled by the work by Per Bak and colleagues[10], which introduced the notion of self-organized criticality (SOC). It was shown that interactive dissipative dynamical systems, such as the Earth's crust, evolve spontaneously (i.e., self-organize) into statistically stationary critical states (i.e., featured by long-range correlations, peculiar power-law scaling behaviors, and divergence of ruling parameters described by universal exponents). Hence, in such physical systems, large-scale instabilities can arise because of the action of stress perturbations regardless of their initial amplitude and depending on the proximity to the critical point. Since then, several studies have interpreted seismicity and tectonic deformations as the results of a dynamical process exhibiting a self-organized critical behavior[1,11–14]. Interestingly, earthquakes in SOC models are clustered in space and time[15], which is a major manifestation of long-term correlation in the seismic process[16].

Nowadays, it is widely accepted by the seismological community that fault interaction contributes to earthquake clustering both in the short- and long-term via stress triggering[17–19] even though seismic activity is far from criticality[20]. According to this, regions of recent high seismic activity have a larger chance of producing new strong earthquakes, in agreement with the clustered occurrence of both historical and recent major seismic events (e.g., the New Madrid 1811–1812 and Central Italy 2016–2017 sequences) whereas regions showing seismic quiescence periods will tend to persist in a quiet state[16,21–23]. This is reflected in the probability distribution of the number of earthquakes over time, which tends to be over-dispersed[24,25].

The previous observations contradict the hypothesis at the foundation of quasi-periodic models for which a large earthquake reduces stress on the rupture zone below that required for additional strong events[26], thus invalidating the characteristic earthquake model[27,28] originally proposed by Schwartz and Coppersmith[29]. Likewise, treating earthquakes as a renewal process[30–32] appears inappropriate by a modeling point of view, as it is assumed that the occurrence of a future seismic event only depends on the previous one and that the probability of occurrence drops to zero immediately after the event. Moreover, the widespread evidence of long-term earthquake clustering invalidates the assumption of seismicity as a Poisson process, which is primarily used in long-term earthquake forecasting (usually known as probabilistic seismic hazard analysis) but it is also applied to produce short-term forecasts[33–35]. Therefore, "*we need a new paradigm of earthquake occurrence*"[28] that incorporates the long memory feature in the seismic process not to disregard the significant statistical correlation between the present and past observations, even across large time shifts.

Although long-term clustering of seismic activity has been recognized and investigated during the past decades[36–38], the first notable attempt to

[1]Dipartimento di Scienze della Terra dell'Ambiente e della Vita, Università degli Studi di Genova, Genova, Italy. [2]Istituto Nazionale di Geofisica e Vulcanologia, Roma, Italy. [3]Institute of Risk Analysis, Prediction and Management (Risks-X), Academy for Advanced Interdisciplinary Studies, Southern University of Science and Technology (SUSTech), Shenzhen, Guangdong, China. [4]Earth Observatory of Singapore, Nanyang Technological University, Singapore, Singapore.
✉e-mail: matteo.taroni@ingv.it

model it was made by Salditch et al.[39], who proposed a new conceptual framework for earthquake occurrence, called Long-Term Fault Memory model. The related probability modeling has been developed by Neely et al.[40]. It assumes that fault regions can be modeled as autonomous physical systems with strain-dependent dynamic behavior (i.e., the probability of large seismic events depends on the accumulated elastic energy potential). The strain change after an event is modeled as a Markov process, so that long-term fault memory emerges due to incomplete strain drop. The final goal of the Long-Term Fault Memory model is a better reconstruction of major paleoseismic records in intraplate fault systems. Hence, it does not take into account the occurrence of events of various magnitudes but just the happening of major seismic sequences as a whole, also neglecting their internal structure (i.e., the triggering of multiple mainshocks during a seismic sequence is equivalent to just a larger earthquake).

To date, a compelling explanation relating long-lasting correlation effects in seismicity to a clear physical mechanism has yet to be provided. Thus, a simple stochastic model for understanding how long-term correlation effects arise in earthquake catalogs is shown in the present article for the first time.

Following in the footsteps of the model proposed by Salditch et al.[39], we present an earthquake catalog simulator that accounts for different earthquake magnitudes through the well-known Gutenberg-Richter relationship. Like its predecessor, our simulator generates time series based on stochastic simulations. Hence, unlike other earthquake simulators[41–47], it does not attempt to reproduce the mechanical aspects of fault rupturing and interaction through the introduction of body forces and application of the constitutive laws of linear elasticity. Although our simulator is not intended to reproduce a purely mechanical modeling of faults, its behavior strongly depends on mechanical concepts—such as energy accumulation, instability thresholds, and discharge imbalances—to capture key features of seismicity like clustering and long memory.

To develop the simulator, we have taken inspiration from the original hydrological problem by Hurst[48], which consists in evaluating the capacity of an ideal water reservoir based on a record of observed discharges from a lake (for details, see Method section). This approach allows us to implement simple physical constraints in our stochastic model and to develop a self-consistent framework for earthquake recurrence that is consistent with the physics of seismicity well beyond previous oversimplified attempts. Briefly, our simulator works as follows:1) the system accumulates strain energy at a specified rate, 2) at a given time, defined according to either an exponential distribution of inter-event times (as in a Poisson process) or corresponding to system instability (which occurs when the system reaches the maximum storable energy), an earthquake occurs with a magnitude sampled from the (truncated) Gutenberg-Richter distribution, and 3) the energy in the system is updated. The process repeats until reaching the desired length of the time series. A modified version of the simulator that reproduces an Omori-like attenuation of seismicity after mainshocks is presented in the Supplementary Information document (Supplementary Note S3), although it has no impact on the main conclusions of the work.

Results

Earthquake simulation

In the following, we generate several synthetic earthquake catalogs by varying the values of the input parameters of the simulator, with the aim of examining their impact on clustering and Hurst's exponent (H), an index of long memory in time series (see Method section). This allows us to analyze the sensitivity of known seismicity features to the loading rate ($\dot{M}_0$), outflow rate (i.e., rate of earthquakes λ_m released by the reservoir), and size of outflowing events. The latter depends on the b value of the Gutenberg-Richter law, minimum magnitude cutoff m_c, and size of the hypothetical seismic reservoir (which is implicitly defined by $m_{\max}$). As with Barani et al.[8], to compute the value of the Hurst exponent, we apply the rescaled range (R/S) analysis[48–52] (see Method section) to time series expressing the number of earthquakes per unit time (e.g., year). Specifically, time series with duration of 2E + 16 years are generated and then analyzed.

Figure 1 shows example time series of M_0 and the associated earthquake catalogs (i.e., moment magnitude versus time) obtained for decreasing values of the loading rate $\dot{M}_0$ (i.e., 2E + 17, 1.5E + 17, 1E + 17 Nm/yr) while keeping the values of the other input parameters fixed: $\lambda_m = 1.3/\text{yr}$, $b = 1$, $m_c = 4$, $m_{\max} = 7.5$. These latter values as well as a loading rate of 2E + 17 Nm/yr are compatible with the seismicity scenario that characterizes the Central Apennines, especially the region struck by the 2009 L'Aquila earthquake[53,54] and the 2016 Amatrice-Visso-Norcia sequence[55,56]. This set of values is referred to as "reference trial" in Table 1, which summarizes the results of the sensitivity analysis. Comparing the panels in Fig. 1, it is immediately evident that decreasing $\dot{M}_0$ results in a decrease in the degree of long memory, with values of H that drops from 0.8 to 0.5 (Fig. S1 in the Supplementary Information). Higher values of the loading rate produce a marked clustering effect (Fig. 1a), which tends to decrease as $\dot{M}_0$ decreases (Fig. 1b), until reaching a random behavior (Fig. 1c) compatible with a Poisson process. Both clustering and random behavior occur because of the disproportion between the loading rate and outflow rate. On the one hand, completely random behavior occurs when the system discharges the accumulated energy at a rate sufficient to avoid reaching the threshold stress (i.e., the accumulated energy never reaches the maximum seismic energy $E_{\max}$ that can be stored in the seismic reservoir). This implies that the results in Fig. 1c are compatible with unpredictable earthquake behavior (i.e., Poisson behavior) and lack of precursory signals[57]. On the other hand, clustering effects (Fig. 1a, b) occur because the system is unable to adequately discharge the accumulated energy, thus reaching the threshold stress. Thus, periods of high release of seismic deformation are followed by years of higher-than-average seismic activity. This is reflected in the frequency distribution of the inter-event time (Fig. 2a, b), which shows a faster decay (especially Fig. 2a) than the distribution associated with a random process (Fig. 2c). This is due to a general acceleration of seismic release near the threshold stress (see Eq. (11)), which therefore implies a higher chance for the occurrence of earthquakes. The duration of a (local) cluster depends on the occurrence of very strong earthquakes (i.e., with magnitude close to $m_{\max}$), which are responsible for a significant drop in the accumulated seismic moment.

Table 1 summarizes the values of H resulting from a set of simulations during which we varied one parameter at a time. For each simulation (trial), the table also displays the values of all input parameters. The value of H associated with each trial should be compared and contrasted with that obtained for the reference simulation, which is displayed in the first row below the header line. As expected, increasing the outflow rate (i.e., λ_m) tends to decrease the value of H. A similar effect is produced by a decrease in the b value, because of the larger proportion of strong earthquakes, which implies a greater release of seismic energy. The same occurs if the minimum magnitude cutoff m_c increases (while keeping λ_m constant). Finally, H decreases as $m_{\max}$ increases due to the concurrent growth of the maximum storable energy. Contrary to expectations, a decrease in λ_m (trial 2a), m_c (trial 4a), or $m_{\max}$ (trial 5a), or an increase in the b value (trial 3a) does not lead to increased H values. Except for trial 5a, for which we observe a slight decrease in H, the values of the Hurst exponent corresponding to trials 2a, 3a, and 4a do not differ substantially from the value associated with the reference simulation. Whereas the decrease in H produced by lowering $m_{\max}$ is attributable to the relative proportion between the (decreased) size of the reservoir and release rate of stronger earthquakes (we recall that $m_{\max}$ has a dual effect, on both the size of the reservoir and distribution of earthquake magnitude), its invariance to a decrease in λ_m and m_c, or to an increase in the b value can be related to a crossover from persistent to independent (or nearly independent) random behavior at large time lags. This effect, which is mild in the cases mentioned above, is enhanced in the example shown in the Supplementary Information (Fig. S2). Also shown in the supplemental material are the magnitude-frequency distributions (Fig. S3) relative to the simulations shown in Fig. 1. Moreover, we also show that incorporating aftershock triggering according to the Omori-Utsu law[58,59] into simulations does not affect long memory.

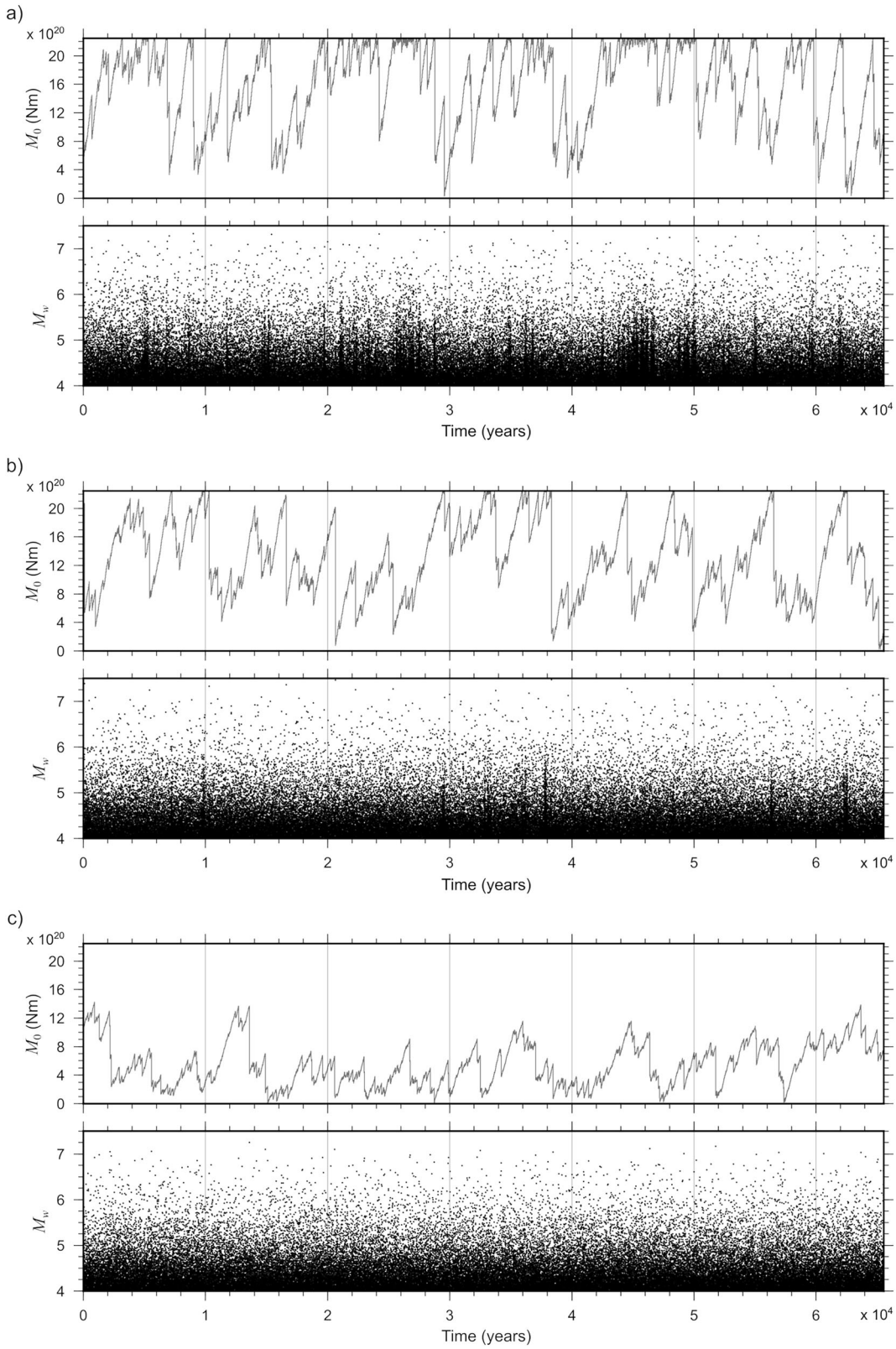

Fig. 1 | Accumulation and release of seismic moment and corresponding earthquake catalog (i.e., simulated magnitudes versus time) for three example simulations with different values of loading rate $\dot{M}_0$ (see Table 1 for full parameterization). a reference case, $\dot{M}_0 = 2.0\text{E} + 17$ Nm/yr; **b** trial 1a, $\dot{M}_0 = 1.5\text{E} + 17$ Nm/yr; **c** trial 1b, $\dot{M}_0 = 1.0\text{E} + 17$ Nm/yr.

Comparisons with Poisson, ETAS, and strain release models

To better understand the behavior of our simulator, we compare and contrast the values of the H exponent and the inter-event time distribution with those resulting from simulations based on two very popular seismicity models: the classical Poisson model, and the ETAS model[60]. The Poisson model has exponential inter-event times that are independent of the magnitude of the earthquakes. The ETAS model instead is used to model the clustering of seismicity, and it uses the Omori law[58,59] to generate seismic

Table 1 | Sensitivity of the Hurst exponent (H) to the input parameters of the simulator: moment rate ($\dot{M}_0$), annual (release) rate of earthquakes (λ_m), b value of the Gutenberg and Richter relationship, minimum magnitude cutoff (m_c), and maximum earthquake magnitude (m_{max})

Trial	Inflow	Outflow				H
	$\dot{M}_0$ (Nm/yr)	λ_m (1/yr)	b	m_c	m_{max}	
Reference	2.0E + 17	1.3	1.0	4.0	7.5	0.80 ± 0.02
1a	1.5E + 17	1.3	1.0	4.0	7.5	0.70 ± 0.02
1b	1.0E + 17	1.3	1.0	4.0	7.5	0.50 ± 0.00
2a	2.0E + 17	0.8	1.0	4.0	7.5	0.79 ± 0.01
2b	2.0E + 17	1.8	1.0	4.0	7.5	0.67 ± 0.02
3a	2.0E + 17	1.3	1.2	4.0	7.5	0.79 ± 0.01
3b	2.0E + 17	1.3	0.8	4.0	7.5	0.49 ± 0.01
4a	2.0E + 17	1.3	1.0	3.5	7.5	0.79 ± 0.01
4b	2.0E + 17	1.3	1.0	4.5	7.5	0.53 ± 0.00
5a	2.0E + 17	1.3	1.0	4.0	7.2	0.74 ± 0.01
5b	2.0E + 17	1.3	1.0	4.0	7.8	0.64 ± 0.02

The values of H were adjusted by applying the correction proposed by Peters[86] to avoid overestimation.

sequences. According to this model, each earthquake can trigger other earthquakes, and stronger events tend to cluster within an "aftershock" sequence. A comparison with the Accelerating Seismic Release (ASR) simulator[61], which is based on the original stress release model of Vere-Jones[62] is also presented and discussed. Except for the energy threshold, this simulator shares some similarities in the loading/unloading process with the one proposed in the present work (as with our simulator, the stress increases linearly with time according to a given rate and decreases when an event occurs).

We generated four very long synthetic catalogs (2^{20} years in length) assuming for all models a truncated Gutenberg-Richter magnitude-frequency distribution with $b = 1, m_c = 4$ and $m_{max} = 7.5$ (we recall that, in the case of our simulator, truncation depends on the amount of energy stored according to Eq. (10)), and a seismic rate of 1.3 events per year, consistently with the parameterization of the "reference trial" in Table 1. The values of the parameters of the ETAS model used here are the same as those in Petrillo & Lippiello[63] without the incorporation of short-term aftershock incompleteness.

The value of the H exponent associated with the catalogs resulting from the three simulators varies substantially. It is 0.72 for the catalog obtained with the simulator presented here, whereas it is around 0.5 for those simulated by using the Poisson, ETAS, and ASR models, thus showing that the latter are unable to mimic long-term correlation in seismicity. However, the ASR simulator requires further investigation. Indeed, for shorter catalogs ($< 2^{14}$ years in length), it might return H values consistent with (moderate) persistent dynamics, although large stochastics fluctuations are observed ($0.5 < H < 0.7$). This contrasting behavior is due to the combined, but competing, effect of the intensity (or risk) function (Eqs. 3 and 4 in Wang et al.[64]), which is assumed to be an exponential function of the stress level (which, in turn, depends on earthquake magnitude), and the truncation of the Gutenberg-Richter law. On the one hand, large magnitudes (i.e., stress levels) lead to a significant increase in risk level (i.e., in the probability of earthquake occurrence in a given time interval) and, locally, to a clustering effect (i.e., growth in long-range correlation). This is consistent with "intermittent criticality" models for which a fault system alternately approaches and retreats from a critical state. On the other hand, due to the truncation of the magnitude-frequency distribution, they have a low chance of being sampled in the simulations. Hence, an increase/decrease in H is expected if a large magnitude event is/is not sampled, thus leading to variable values of the Hurst exponent (especially evident in short catalogs). Conversely, our simulator allows generating catalogs characterized by $H > 0.5$ regardless of their length. Indeed, it uses a global parameter (i.e., stored energy) to control the inter-event time distribution of seismic events within a crustal volume. The deep conceptual difference between TaBaTa and the ASR simulator (energy budget-based clustering versus event-related clustering) shows that, while the ASR model is able to reproduce long-range correlation effects only locally, our model allows a generalization of long-term persistence even in the long run, in agreement with paleoseismic observations.

Hereafter, we will focus on the comparison with the more popular Poisson and ETAS models.

Figure 3 shows the inter-event time (cumulative) distributions for very large events (i.e., $M \geq 7$) and the joint estimation of β (i.e., $b/1.5$) and the corner magnitude of the magnitude-frequency distributions resulting from the simulations[65]. The departure between the inter-event time distributions is tested through the two-sample Kolmogorov-Smirnov test[66,67]. Moreover, we also investigate the correlation between earthquake magnitude and the inter-event time of subsequent events by computing the conventional Pearson correlation coefficient along with its significance[68]. The results of the tests are summarized in Table 2 for $M \geq 7$ and $M \geq 5$. For the case $M \geq 7$, comparing the inter-event time distributions reveals that the one obtained with our simulator is significantly different from those resulting from the Poisson and ETAS models, which instead are similar to each other. This similarity is related to the magnitudes considered in this test (i.e., $M \geq 7$). It is known, indeed, that the triggering component of the ETAS model has a stronger effect when one focuses on smaller events (note the very small but significant negative correlation between magnitude and the inter-event time of subsequent events obtained for $M \geq 5$). Our simulator tends to produce a large percentage of longer inter-event times, which is reflected in a small but significant positive correlation between magnitude and the inter-event time relative to a subsequent event. Such a positive correlation decreases substantially when smaller earthquakes are included in the analysis (see Table 2 for $M \geq 5$). In other words, very strong earthquakes generated by our simulator are less frequent than those obtained via the conventional ETAS and Poisson models. This is reflected in the smaller value of the corner magnitude of the magnitude-frequency distribution (Fig. 3b).

In addition to the previous analyses, for the case $M \geq 7$, we also investigated the burstiness[69] of the simulated seismicity, $B = (\sigma - \mu)/(\sigma + \mu)$, where μ is the mean of the inter-event times and σ is their standard deviation. Such a coefficient was deeply examined by Griffin et al.[70]. By analyzing a global compilation of paleoseismic records, they observe that seismicity on individual fault segments exhibits a negative value of B, implying that large earthquakes recur more regularly than a random Poisson process. Analyzing the burstiness of the catalogs simulated in the present study shows that both the data sets generated with the ETAS and Poisson simulators are featured by null burstiness (i.e., seismicity occurs

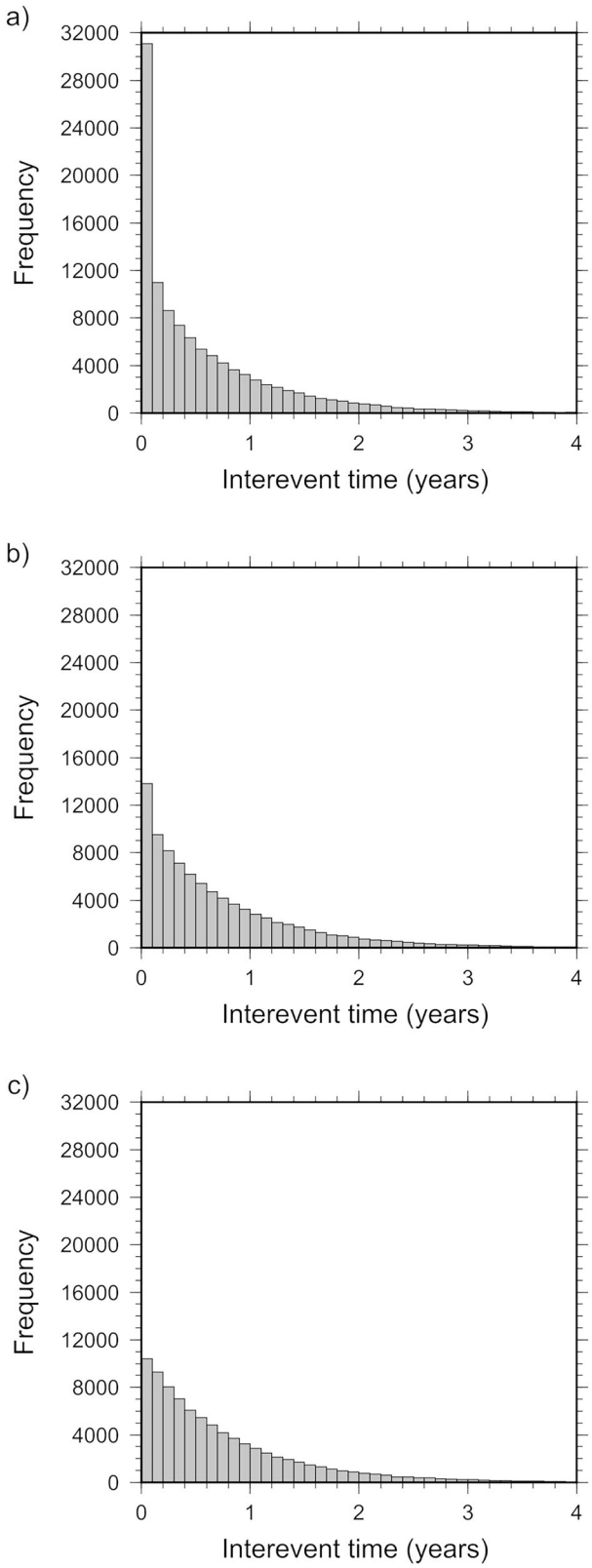

Fig. 2 | Frequency distribution of the inter-event time for the three examples presented in Fig. 1 (see also Table 1). a reference case; **b** trial 1a; **c** trial 1b.

essentially randomly over time). Conversely, B is significantly different from zero, $B = -0.14 \pm 0.01$ (the uncertainty has been estimated with both analytical error propagation and bootstrapping), for the long memory catalog generated with our simulator, thus proving that it is able to replicate observed features of real seismicity.

The previous findings are completely in agreement with the physical foundations of our simulator, especially for large-magnitude earthquakes. Specifically, on a single fault, a strong event (i.e., with magnitude close to $m_{\max}$) releases a large amount of the energy stored in the system and, consequently, a longer inter-event time is expected until the next large earthquake. Complex patterns and persistence are instead determined by partial ruptures (in agreement with Barbot[71]) producing a fan of long-range interactions between different fault patches whose dynamic evolution is ultimately controlled, in our view, by the statistical properties of the potential energy profile of the fault system.

Discussion and conclusions

Physical interpretation

The analogy between fluid flows and seismic occurrence as a process of strain energy accumulation and release (see the paragraph "Earthquake catalog simulator") requires physical justification and detailed interpretation. In this paragraph, we provide motivation and physical soundness of our model. While the implementation of fundamental statistical laws (i.e., the Gutenberg-Richter law and the exponential distribution of inter-event times) does not require special explanation, the existence of an upper bound for strain energy accumulation forcing the occurrence of seismic events, according to the assumptions behind our simulation, must be justified.

Earthquakes are dynamical instabilities described by fracture mechanics. Their onset is controlled by the mechanical strength of structural asperities on the fault, preventing slip along otherwise weak interfaces (Fig. 4a, b). Hence, seismic events can be modeled as dynamically unstable ruptures whose spatial and temporal evolution depends on rock heterogeneities acting as amplifiers or, conversely, barriers for the mobilization of crustal volumes. Then, although strain deformation progressively accumulates uniformly due to the action of tectonic forces, heterogeneities produce extremely variable strain (and stress) concentrations (Fig. 4c). The Weibull distribution is usually used to model the strength of materials to rupture and fault resistance to slip[72–74] (Fig. 4d). While the average stress on fault increases, a growing number of asperities reach their yielding point, triggering cascades of ruptures. Simulations can be performed using the approach described in Zaccagnino et al.[75].

Let $\langle\sigma\rangle_t$ be the average stress release on a fault due to the seismic events occurring on it, defined as:

$$\langle\sigma\rangle_t \sim R\overline{\triangle\sigma}(t) \tag{1}$$

where R here represents the seismic rate (i.e., frequency) of events and $\overline{\triangle\sigma}(t)$ is the mean stress drop at time t produced by the occurred earthquakes. The shear stress, σ_s, is a good approximation of the stress drop, $\triangle\sigma$, for large earthquakes[76].

According to Dieterich[77], the seismic rate due to triggered events increases roughly exponentially as a function of the shear stress, σ_s, close to the mainshock:

$$R = ce^{\frac{\triangle\sigma}{A\sigma_n}} \propto re^{\alpha\sigma_s} \tag{2}$$

where σ_n represents the normal stress and α, c, A, and r are constant. This condition is expected to occur in our model in the proximity of the maximum possible energy accumulated in the system ($E_{\max}$), which corresponds to the threshold stress, σ_c.

In turn, the mean stress drop can also be expressed in statistical terms at the time of large earthquakes ($t = t_c$) as

$$\overline{\triangle\sigma}(t = t_c) \sim \frac{\lambda}{k}\Gamma\left(\frac{1}{k}\right) \pm \frac{\lambda}{k}\sqrt{2k\Gamma\left(\frac{2}{k}\right) - \Gamma^2\left(\frac{1}{k}\right)} \tag{3}$$

where k and λ are the shape and scale parameters of the Weibull distribution of the strength on fault and Γ is the Gamma function.

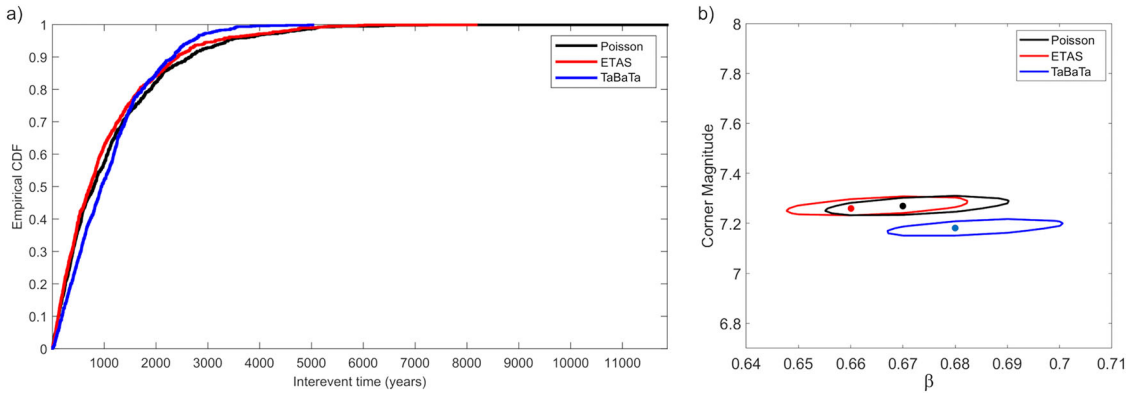

Fig. 3 | Inter-event time distributions and joint estimation of the parameters of the magnitude-frequency distributions obtained from the simulations. a Inter-event time (cumulative) distributions resulting from the simulations based on the Poisson model, ETAS model, and the simulator presented in this work for $M \geq 7$ events. **b** Maximum likelihood estimation of the β ($= b/1.5$) coefficient and corner magnitude (dots) of the tapered Gutenberg-Richter distribution and relative 95% confidence region (ellipses) for the same three models mentioned above.

Table 2 | Correlation (measured by Pearson's correlation coefficient) between magnitude and the inter-event time of subsequent events (lines 2–4) and departure (measured by the *p*-value associated with the Kolmogorov-Smirnov test) between the empirical (cumulative) distributions of the inter-event time for each pair of catalogs generated by using the simulator present here (TaBaTa), the Poisson model, and the ETAS model (lines 5–7)

Catalog	$M \geq 7$			$M \geq 5$		
	Number of events	Correlation Coefficient	*p*-value	Number of events	Correlation Coefficient	*p*-value
Poisson	915	0.014	0.67	136110	0.0007	0.79
ETAS	996	0.003	0.93	143557	-0.01	**$1.4\ 10^{-4}$**
TaBaTa	946	0.162	**$5.3\ 10^{-7}$**	168870	0.06	**$<10^{-100}$**
Poisson vs. TaBaTa	-	-	**$1.8\ 10^{-4}$**	-	-	**$<10^{-100}$**
Poisson vs. ETAS	-	-	0.10	-	-	**$1.7\ 10^{-27}$**
ETAS vs. TaBaTa	-	-	**$1.7\ 10^{-7}$**	-	-	**$<10^{-100}$**

Earthquakes with $M \geq 7$ and $M \geq 5$ are considered. Bold text indicates significant *p*-values (i.e., *p*-value < 0.01/6 where 6 is the number of correlations tested according to the Bonferroni correction for multiple tests[87]).

As stated above, there exists a transition from a globally stable configuration with isolated asperities breaking sub-critically to a widespread instability allowing large-scale destabilization on the fault with complete stress drop on it, which roughly corresponds to the stress condition at which $\langle\sigma\rangle_t$ reaches the value of the accumulated tectonic stress since the last large event $\langle\sigma\rangle_g = \dot{\sigma}_g t$ (see Fig. 4e), namely:

$$\sigma_c = \langle\sigma\rangle_{t_c} = \sigma_s(t = t_c) = \dot{\sigma}_g t_c \sim \frac{\lambda r}{k}\Gamma\left(\frac{1}{k}\right)e^{\frac{\dot{\sigma}_g t_c}{\lambda \sigma_n}} \quad (4)$$

Therefore, assuming a progressive breakdown of the system due to the action of tectonic stress, the average value of shear stress on the fault is expected to drop swiftly approaching the threshold stress σ_c (see Fig. 4f), according to:

$$\sigma_s(t) \sim \frac{\sigma_0}{2}\left(1 + \tanh\left(\frac{\sigma_c - \langle\sigma\rangle_t}{W(k,\lambda)}\right)\right) \quad (5)$$

being σ_0 a constant and W a parameter controlled by the statistical properties of the fault strength. Barring stochastic fluctuations (for strain and energy potential), the stress σ_c can be considered as an upper bound for stress accumulation on a structurally homogeneous fault. Thus, the hypothesis b) of our model (see the paragraph "Earthquake catalog simulator") is physically justified.

In real fault systems, energy accumulation and fault locking are promoted by structurally complex, segmented, or immature fault systems characterized by higher fractal dimension[78], often in slowly deforming continental regions, as already discussed in the introduction. Conversely, relatively simpler fault networks with a few misaligned seismogenetic structures and poorly competent rocks (i.e., creeping segments) are usually less prone to host clustered seismicity[79]. Then, our model may be of special interest for the simulation and reproduction of seismic activity in complex tectonic regions (e.g., the Basin and Range province, Apennines, continental Greece and central Europe, Tibet and Türkiye fault systems), especially wherever bursty and clustered seismicity is observed.

Conclusions

In this work, we have presented a simple physics-informed stochastic earthquake simulator based on well-known statistical features of seismicity. It includes a physical constraint on the largest magnitude that can be released at a given time and an upper effective bound for the energy budget of crustal volumes, which is characterized by the occurrence of an avalanche-like dynamic transition between a macroscopically locked fault to a widespread slip. The latter constraint is borrowed from Hurst's model for water reservoirs.

The presence of long-memory depends by the ratio of the loading rate to the rate of energy discharge, being the first due to the joint effect of the long-standing action of tectonic forces and seismic coupling, and the second controlled by avalanche-like dynamics of seismic activity. Figure 1 shows this concept assuming different loading rates while keeping the outflow rate fixed (the same can be observed by comparing the results of Trial 2a and Trial 2b in Table 1, which assume different earthquake release rates while keeping the loading rate fixed). In our model, the system develops "long-memory" while approaching and keeping itself close to the instability condition (i.e., when the energy in the reservoir reaches the maximum storable energy). There is no limitation to the

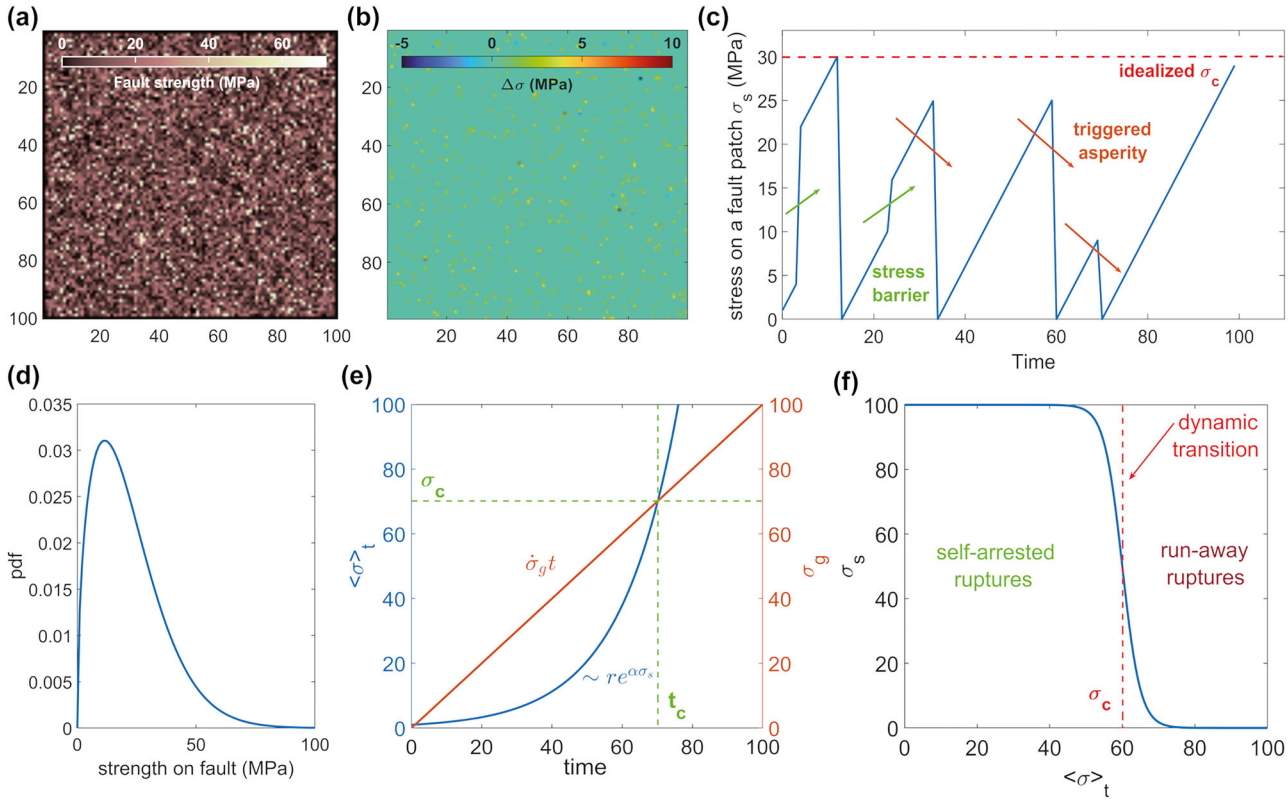

Fig. 4 | Schematic representation of the physical mechanism behind memory on faults. **a** Fault strength distribution; **b** a snapshot of the stress drops on a simulated fault; **c** time evolution of the stress on a specific point on a fault; **d** probability density distribution of strength on fault; **e**,**f** the competition between triggered seismic activity and tectonic stress accumulation determines a stress value of transition between a macroscopically stable fault with self-arresting minor ruptures and run-away events.

outflow rate of stored energy, which is controlled by the frequency distribution of earthquake magnitudes (i.e., by the Gutenberg-Richter relationship). According to the Gutenberg-Richter law, large earthquakes are relatively rare with respect to smaller ones. In other words, they have less chance of being sampled in the simulations. Thus, the system may remain at the edge of instability for a prolonged time, releasing energy through smaller events. This leads to temporal clustering.

Statistical analysis of simulated catalogs has revealed that long-term correlations in seismic activity, which Poisson and ETAS models are unable to capture, may be related to the proximity to such dynamic instability. Specifically, if the system is unable to discharge the accumulated energy at a sufficient rate, a time is reached at which the internal clock of the system speeds up, producing higher chances of major events in order to make the system stable again. This process may be responsible for complex seismicity patterns, with periods of high seismic release that tend to be followed by years of higher-than-average seismicity (thus implying intense clustering). This behavior is reflected in variable (i.e., non-stationary) seismicity rates. Compared to the memoryless Poisson behavior, which can be observed when the release rate of earthquake energy is greater than the loading rate (so that the seismic reservoir never fills up completely), earthquake predictability is, at least in principle, higher. Indeed, when the system approaches the threshold stress, an earthquake is more likely to happen. Conversely, when a very strong event occurs (with magnitude close to the maximum magnitude associated with the seismic reservoir), producing an abrupt change in the system due to a significant drop in the accumulated energy, the probability of another strong event in the near future is reduced. As indicated by the analysis of the burstiness coefficient, such behavior is compatible with worldwide paleoseismic observations[70,80,81]. This behavior is also consistent with recent observations of low-level seismicity in zones of high co-seismic slip associated with very large earthquakes; considerable waiting times are expected for these zones until the next major earthquake[23,82].

Although our simulator is quite simple (e.g., it cannot generate cascading interaction between faults), being based on a few physical assumptions and elementary statistical laws used in seismology, it allows showing that complex seismicity characteristics can be explained by simple rules. Therefore, although the current version is not yet suitable for earthquake forecasting, it lays the groundwork for future developments in this direction. With further development, the model could complement existing approaches by capturing features of seismicity that models like the standard ETAS may overlook, especially in regions where long-term clustering is significant. Lastly, future versions of this simulator should take into account the interaction between different faults.

Method

In this section, we present the catalog simulator along with the underlying assumptions. Special attention is given to the description of Hurst's problem[48], which is at the foundation of the simulator, and to the related rescaled range analysis[48–52].

Hurst's problem and rescaled range analysis

What storage would have been required for a reservoir to release a volume each year equal to the mean influx for a given period? This was the original question that Hurst sought to answer during his career. In Hurst[48], the author tackled the problem of determining the capacity of an ideal water reservoir, which never overflows or empties, based on a record of observed discharges from a lake. In any given year, t, the reservoir accepts the influx $x(t)$ from the lake and a given volume per year (discharge), assumed equal to the average influx over the period of τ years $\left(\text{i.e.}, \bar{x}_\tau = 1/\tau \sum_{t=1}^{\tau} x(t)\right.$ where t is a discrete integer-valued time), is released from it. The difference between the maximum and minimum amounts of water contained in the

reservoir over the time span of interest $R(\tau)$ represents the storage capacity required to maintain the mean discharge throughout the period τ:

$$R(\tau) = \max_{1 \le t \le \tau} X(t,\tau) - \min_{1 \le t \le \tau} X(t,\tau) \tag{6}$$

where $X(t)$ is the accumulated departure of the influx $x(t)$ from $\bar{x}_\tau$:

$$X(t,\tau) = \sum_{j=1}^{t} \left(x(j) - \bar{x}_\tau \right) \quad t = 1, \ldots, \tau \tag{7}$$

Alternatively, the quantity $X(t,\tau)$ can be also computed as:

$$X(t,\tau) = \left(\sum_{j=1}^{t} x(t) \right) - t\bar{x}_\tau \quad t = 1, \ldots, \tau \tag{8}$$

Clearly, the range $R(\tau)$ depends on the time period τ and we expect it to increase as τ increases.

It is worth noting that, compared to the case of the water reservoir just described, in seismological problems $R(\tau)$ provides an estimate of the total elastic energy that can be released by a hypothetical seismic reservoir (i.e., a fault or, more generally, a volume of the Earth's crust within which earthquake energy accumulates) over a certain time period τ rather than an estimate of the storable energy[6,52]. Indeed, seismologists do not measure stress loading (i.e., influx) but the energy released by earthquakes (or the number of earthquakes released in a certain crustal volume) with time.

In order to compare the range $R(\tau)$ of different natural phenomena excluding the effect of τ[50,51], Hurst introduced the dimensionless R/S ratio (the so-called rescaled range), where S is the standard deviation of $x(t)$ over the period τ. It was observed that:

$$\frac{R(\tau)}{S(\tau)} = c\tau^H \tag{9}$$

where H, an index of long memory of time series, is the so-called Hurst exponent. Specifically, H lies between 0 and 1 and indicates the following properties according to the value assumed:

- $H > 0.5$ indicates a stationary, long memory process with persistent behavior, meaning that a high value in a time series will tend to be followed by another high value and, conversely, that a decrease in values will most likely be followed by a decrease; as H becomes closer to one, the likelihood of observing large and abrupt changes in the series increases[83,84].
- $H < 0.5$ indicates an anti-persistent behavior, meaning that an increase in a time series will most likely be followed by a decrease or vice-versa.
- $H = 0.5$ indicates a completely uncorrelated time series (e.g., random walk, or a Poisson process).

The previous classification depends on the speed of growth of the R/S ratio. A random system does not produce coherent evolutions; thus, R increases linearly while the standard deviation follows a square-root-like behavior, so that their ratio shows a diffusive trend. Persistent trends show faster growth. Conversely, fluctuating dynamics with anti-persistent, destructive patterns are characterized by a slowly growing or even constant ratio.

Computationally, the value of H can be easily obtained via regression analysis. The time series at hand is divided into a number N of (non-overlapping) windows of size τ, the rescaled range $R(\tau)/S(\tau)$ is computed for each of them, and the obtained values are averaged to get the final R/S ratio for that value of τ. The procedure is repeated considering time windows of increased size (typically, following a power-of-two scheme, so that $N = 2^0$, 2^1, 2^2…, 2^h with h usually not less than 5), till reaching the entire length of the time series. Regression of $R(\tau)/S(\tau)$ versus τ leads to the value of H.

Earthquake catalog simulator

To understand the long memory effects in earthquake catalogs, we have taken inspiration from the inflow-outflow process described by Hurst (Fig. 5). The analogy is based on the similarities between the long-term recordings of floods and water levels in reservoirs and the elastic energy accumulation and release in the brittle crust. Unlike the case of the water reservoir for which the influx time series is known, seismologists deal with time series of "outflowing" energy (i.e., energy released by earthquakes over time). In a sense, this flips the original problem, especially the definition of inflow and outflow; namely, we assume that every year the seismic reservoir will release a certain amount of energy $x(t)$, and a given amount of seismic energy per year $\bar{x}_\tau$, assumed equal to the average energy released over τ years, will be accumulated. Therefore, while loaded at a given energy rate, $\bar{x}_\tau$, the system is supposed to release each year a certain amount of energy $x(t)$.

The process of energy accumulation and release is managed via a simple stochastic simulation model that relies on the following basic assumptions:

a) earthquake magnitudes are drawn from the Gutenberg-Richter distribution with slope $\beta = b\ln 10$, minimum magnitude cutoff m_c, and upper bound magnitude (m_u) given by the total energy stored in the reservoir at time t (note that here t no longer denotes a discrete integer-valued time but continuous time); basically, we sample the cumulative distribution function of the truncated Gutenberg-Richter relation[85]:

$$F_M(m) = \frac{1 - e^{-\beta(m - m_c)}}{1 - e^{-\beta(m_u - m_c)}} \quad m_c \le m \le m_u \tag{10}$$

b) inter-event times are drawn from an exponential distribution (i.e., as in a Poisson process) with rate parameter λ_m except when the total accumulated energy reaches the maximum energy ($E_{\max}$) that can be stored in the reservoir. Such an energy corresponds to the maximum magnitude ($m_{\max}$) associated with the fault (or crustal volume). We recall that, according to Hurst[48], the overflow of the reservoir is not allowed. Therefore, whenever the instability condition is reached, an earthquake instantly occurs, and its occurrence time is equal to:

$$t_i = t_{i-1} + \frac{E_{\max} - E_{i-1}}{\dot{E}} \quad i = 1, \ldots, n \tag{11}$$

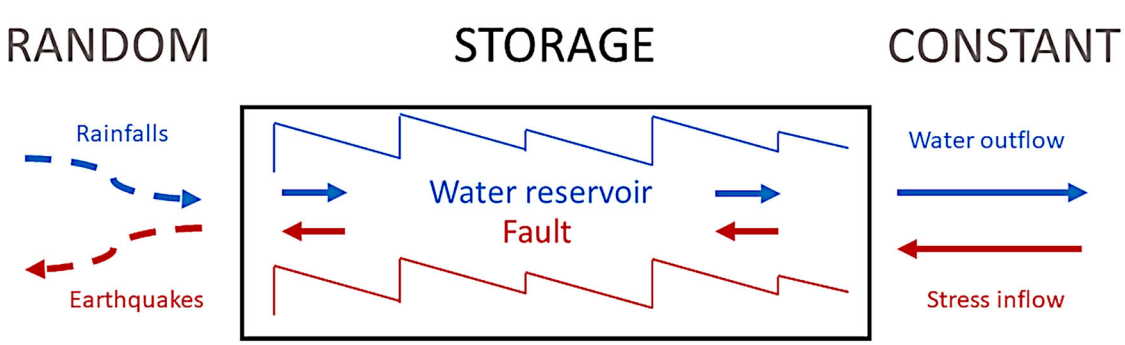

Fig. 5 | Schematic representation of the Hurst problem for water (blue) and seismic (red) reservoirs.

where E_i is the energy stored in the reservoir at the ith time step, and $\dot{E}$ is the energy rate (i.e., loading rate). Equation (11) implies an acceleration in seismic release.

Similarly, if the stored energy falls below the energy corresponding to m_c (E_c), the time to the next earthquake with magnitude m_c or above should not be lower than:

$$t_i = t_{i-1} + \frac{E_c - E_{i-1}}{\dot{E}} \quad i = 1, \ldots, n \tag{12}$$

From a computational perspective, the simulator, called TaBaTa (see paragraph of Data Availability), works as follows:1) the strain energy within a crustal volume is accumulated at given rate $\dot{E}$, 2) at time t_i, defined according to either an exponential distribution of inter-event times or Eq. (11) if the instability condition (i.e., the maximum storable energy) is reached, an earthquake occurs with a magnitude sampled from the (truncated) Gutenberg-Richter distribution (Eq. (10)), and 3) the energy in the system is updated. The process repeats until reaching the end of the time series, whose length is defined in input.

For ease of application, in the computer code, we replace strain energy and the relative rate with seismic moment (M_0) and moment rate ($\dot{M}_0$). The latter quantities, indeed, can be more easily determined from either seismic, geodetic, or geologic data.

Data availability

Data used in our analyses and model output are available at https://doi.org/10.5281/zenodo.15850311

Code availability

TaBaTa.m and TaBaTaOU.m are available at https://doi.org/10.5281/zenodo.15850311 along with other codes used in our analyses.

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Author contributions
S.B. and M.T. conceived the study, coded the simulators (TaBaTa.m and TaBaTaOU.m), and carried out most of the analyses. S.B. wrote the majority of the manuscript with contributions from M.T. and D.Z. D.Z. contributed to the physical interpretation of the model. P.A.H. conceived Fig. 2 and contributed to the Introduction. G.P. coded the ASR simulator with contributions from S.B. and provided the time series based on the ETAS model. All authors contributed to manuscript editing and figure preparations, simulations, and data processing. All authors reviewed the manuscript and contributed to the interpretation of the results.

Competing interests
The authors declare no competing interests.

Additional information
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Correspondence and requests for materials should be addressed to Matteo Taroni.

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